

Targeted SAT Math — Set 03

Student: Karina

Homework after Test 9. Work on paper. Calculator / Desmos is fine.

30 questions. These are only the leaks from Test 9 (and two leftover slope items). Items are instructor-written SAT-style questions, **not** official College Board stems, and **not** taken from Practice Tests 1–11.

SPR = student-produced response. Write a number (integer, decimal, or fraction).

Before you calculate: write a 3–5 word **job** on scratch (**max value, smallest, by how many, sum of volumes**). Then copy numbers.

1. Notes for this set only

Frequency table

- **Data value** = the number in the first column (what was measured).
- **Frequency** = how many times that value occurs. It is a count, not a data value.
- “Maximum **data value**” = the largest number that actually appears in the set. Frequency 0 means that value is **not** in the set.
- Do **not** multiply value $\times$ frequency. That product is not the maximum.

Square-root equations — smallest solution

- If both sides have a square root (or one side does), **square both sides**. Do not stop after looking at one side.
- $\sqrt{(x - h)^2} = |x - h|$. That is not “ $x \geq h$, so the answer is h .”
- A domain leftover ($x \geq 2$, $x \geq 4$) is a restriction, **not** a solution. Solve, then check.
- “Smallest solution” means the lesser number that actually works. Check both candidates in the original equation.

“By how many”

- Scale the sample to the full group **first**.

- Then **subtract**. “By how many more / by how many would A beat B” is a difference, not a total.
- Trap 1 = the sample difference (never scaled). Trap 2 = the scaled total for A only.

Similar solids

- Linear scale k from surface areas: $k^2 = \frac{SA_{\text{large}}}{SA_{\text{small}}}$ as **one fraction**, then $k = \sqrt{\text{that}}$.
- Do **not** take $\sqrt{SA_{\text{large}}}$ and $\sqrt{SA_{\text{small}}}$ as two separate calculator hits. That sprays decimals and you lose the clean square.
- Volumes scale with k^3 . Areas with k^2 .

Adding rational expressions

- If one denominator is $a - b$ and the other is $b - a$, rewrite $b - a = -(a - b)$ **before** you add.
- Do not invent a new free denominator and dump both numerators on top.

“Displays ... as a constant or coefficient”

- The feature they named (maximum, minimum, y -intercept) must sit out as a number you can read — a constant term or a coefficient — with no leftover arithmetic.
- Vertex form displays a max/min as the outside constant. Factored form does not.
- $a \cdot b^x + c$ displays c , not the y -intercept (the intercept is $a + c$).
- $a \cdot b^{x+k}$ rewritten as $(ab^k) b^x$ displays the intercept as the coefficient ab^k .

Leftover (two items only)

- Same y on center and contact point $\Rightarrow$ tangent is **vertical** $\Rightarrow$ slope **undefined**.
- Same $x \Rightarrow$ tangent is **horizontal** $\Rightarrow$ slope **0**.

2. Questions

1. (SPR) The frequency table summarizes a data set.

Data value	Frequency
4	9

Data value	Frequency
5	12
6	8
7	0
8	5
9	11

What is the maximum data value in the data set?

Answer: __

2. (SPR) The frequency table summarizes a data set.

Data value	Frequency
2	4
5	15
8	7
11	2
14	6
17	0
18	3

What is the maximum data value in the data set?

Answer: __

3. The frequency table summarizes a data set.

Data value	Frequency
10	2
20	20
30	8
40	1

What is the maximum data value in the data set?

- A) 20
- B) 40
- C) 400
- D) 61

4. (SPR)

$$\sqrt{x(x-3)} = \sqrt{3x+16}$$

What is the smallest solution to the given equation?

Answer: __

5. (SPR)

$$\sqrt{x(x-5)} = \sqrt{x+27}$$

What is the smallest solution to the given equation?

Answer: __

6. (SPR)

$$\sqrt{x(x-2)} = \sqrt{35}$$

What is the smallest solution to the given equation?

Answer: __

7. (SPR)

$$\sqrt{x(x-6)} = 4$$

What is the smallest solution to the given equation?

Answer: __

8. (SPR)

$$\sqrt{(x-4)^2} = 7$$

What is the smallest solution to the given equation?

Answer: __

9. (SPR)

$$\sqrt{(x-3)^2} = \sqrt{3x+19}$$

What is the smallest solution to the given equation?

Answer: __

10. (SPR)

$$\sqrt{5x+6} = x+2$$

What is the smallest solution to the given equation?

Answer: __

11. A bakery sampled 250 customers at random: 160 bought a muffin and 90 bought a scone. If 2,000 customers buy one of those two items, by how many more muffins than scones are expected?

- A) 70
- B) 560
- C) 1,280
- D) 1,750

12. A warehouse sampled 120 boxes: 75 marked rush and 45 marked standard. If 960 boxes ship this week, by how many more rush boxes than standard boxes are expected?

- A) 30
- B) 240
- C) 600
- D) 840

13. An app store sampled 400 reviews: 260 preferred Game A and 140 preferred Game B. If 5,200 users each pick one of the two games, by how many more users is Game A expected to be chosen?

- A) 120
- B) 1,560
- C) 3,380
- D) 4,800

14. A nursery sampled 90 seedlings: 54 tomato and 36 pepper. If 630 seedlings are planted, by how many more tomato plants than pepper plants are expected?

- A) 18
- B) 126
- C) 378
- D) 540

15. A library sampled 175 hold requests: 105 for mysteries and 70 for science fiction. If 2,100 holds are placed, by how many more mystery holds than science-fiction holds are expected?

- A) 35
- B) 420
- C) 1,260
- D) 1,925

16. (SPR) A transit survey sampled 48 riders: 30 took Route A and 18 took Route B. If 720 riders take one of those two routes, by how many more Route A riders than Route B riders are expected?

Answer: __

17. (SPR) Right circular cone P is similar to right circular cone Q . The surface area of cone P is 48 square centimeters and the surface area of cone Q is 1,200 square centimeters. The volume of cone Q is 2,000 cubic centimeters. What is the sum of the volumes, in cubic centimeters, of cone P and cone Q ?

Answer: __

18. (SPR) Right cylinder A is similar to right cylinder B . The surface area of cylinder A is 75 square inches and the surface area of cylinder B is 1,200 square inches. The volume of cylinder B is 1,024 cubic inches. What is the sum of the volumes, in cubic inches, of cylinder A and cylinder B ?

Answer: __

19. (SPR) Pyramid R is similar to pyramid S . The surface area of pyramid R is 32 square meters and the surface area of pyramid S is 800 square meters. The volume of pyramid R is 6 cubic meters. What is the volume, in cubic meters, of pyramid S ?

Answer: __

20. (SPR) Right rectangular prism M is similar to right rectangular prism N . The surface area of prism M is 54 square centimeters and the surface area of prism N is 1,350 square centimeters. The volume of prism N is 1,000 cubic centimeters. What is the sum of the volumes, in cubic centimeters, of prism M and prism N ?

Answer: __

21. Which expression is equivalent to $\frac{4y}{x-y} + \frac{4x}{y-x}$?

- A) 4
- B) -4

- C) $\frac{4(x+y)}{x-y}$
D) $\frac{8xy}{x-y}$

22. Which expression is equivalent to $\frac{m}{m-n} + \frac{n}{n-m}$?

- A) 0
B) 1
C) $\frac{m+n}{m-n}$
D) $\frac{m-n}{n-m}$

23. Which expression is equivalent to $\frac{2a}{a-b} - \frac{2b}{b-a}$?

- A) 2
B) $\frac{2(a-b)}{a+b}$
C) $\frac{2(a+b)}{a-b}$
D) $\frac{2(a-b)}{b-a}$

24. The function f is defined by $f(x) = -3(x+1)(x-5)$. Which of the following equivalent forms displays the maximum value of f as a constant or coefficient?

- I. $f(x) = -3(x-2)^2 + 27$
II. $f(x) = -3x^2 + 12x + 15$

- A) I only
B) II only
C) I and II
D) Neither I nor II

25. The function f is defined by $f(x) = 2(x-4)^2 + 9$. Which of the following equivalent forms displays the minimum value of f as a constant or coefficient?

- I. $f(x) = 2(x-4)^2 + 9$
II. $f(x) = 2x^2 - 16x + 41$

- A) I only
B) II only
C) I and II
D) Neither I nor II

26. The function f is defined by $f(x) = 5(1.4)^{x+2}$. Which of the following equivalent forms displays the y -coordinate of the y -intercept of the graph of $y = f(x)$ as a constant or coefficient?

I. $g(x) = 9.8(1.4)^x$

II. $h(x) = 5(1.4)^x + 2$

A) I only

B) II only

C) I and II

D) Neither I nor II

27. The function f is defined by $f(x) = 3(1.5)^x + 7$. Which of the following equivalent forms displays the y -coordinate of the y -intercept of the graph of $y = f(x)$ as a constant or coefficient?

I. $g(x) = 3(1.5)^x + 7$

II. $h(x) = 10 + 3((1.5)^x - 1)$

A) I only

B) II only

C) I and II

D) Neither I nor II

28. A circle has center $(3, 8)$. A line is tangent to the circle at $(11, 8)$. What is the slope of the tangent line?

A) 0

B) $\frac{8}{11}$

C) undefined

D) 8

29. A circle has center $(6, 1)$. A line is tangent to the circle at $(6, 9)$. What is the slope of the tangent line?

A) 0

B) undefined

C) 8

D) $\frac{1}{6}$

30. (SPR) A theater surveyed 80 ticket buyers: 50 chose the matinee and 30 chose the evening show. If 640 tickets are sold, by how many more matinee tickets than evening tickets are expected?

Answer: ____